

Test Data Generation Report - sample-qa-test-project

Deterministic staging validation run

Run date: Aug 5, 2026

Run identifier: sample-qa-test-project-20260805-113952

Requested branch: agent/qa-validation-fixture

Exact commit: 84d392ec8f0d30da10fccdbf00ea2bcfae3922c5

Repository: https://github.com/kashyapempiricinfotech-sys/sample-qa-test-project

RUN STATUS: COMPLETED WITH FINDINGS — staging identity verified; all 9 forms executed; 6 retested gaps filed.

Executive Summary

The requested branch was resolved to the exact commit above and analyzed in full. The repository defines nine in-scope form surfaces. Because the count is below the 25-form cap, all nine were selected for execution. A fresh disposable staging fixture was started and its environment endpoint confirmed staging, production=false, sample_only=true, seed version qa-oracle-v2, and last_seeded_at 2026-08-05T06:06:23.015711+00:00.

The retry reached the loopback staging fixture in the in-app browser. All nine in-scope forms were executed. Seventeen run-tagged records were created with intact relationships and zero orphans. Six validation gaps were reproduced with an identical safe retest and filed as Jira Bugs: Critical 1, High 3, Medium 1, and Low 1. GAP-03 remains manual-review only because submitting a nonexistent product would intentionally create an orphan. CONTROL-01 passed on its identical retry and is not a defect.

Measure	Result
Forms discovered	9
Forms selected	9 (all; cap not exceeded)
Forms browser-tested	9
Forms skipped due to cap	0
Dataset rows generated	103
Records created in staging	17
Confirmed validation gaps	6 - Critical 1, High 3, Medium 1, Low 1
Jira bugs created	6

Repository and Branch/Commit Analyzed

Item	Recorded value
Repository	https://github.com/kashyapempiricinfotech-sys/sample-qa-test-project
Requested branch	agent/qa-validation-fixture
Exact commit SHA	84d392ec8f0d30da10fccdbf00ea2bcfae3922c5
Commit message	Add deterministic QA staging validation oracle

Item	Recorded value
Staging base URL declared in repository	http://127.0.0.1:8000/staging
Environment endpoint	http://127.0.0.1:8000/api/environment
Fixture version	qa-oracle-v2

Analysis covered application routes, HTML templates, Pydantic schemas, SQLAlchemy models, migrations, deterministic fixtures/factories, the machine-readable intentional-gap oracle, startup/reset scripts, and the repository README. The hidden legacy contact route was excluded because it is explicitly deprecated and absent from navigation.

Forms Discovered and Forms Tested

#	Form	Route	Surface	Execution status
1	Staging login	/staging/login	Login page	Tested — successful staging login
2	Dashboard Quick note	/staging	Same-URL client tab	Tested in browser
3	Registration	/staging/forms/registration	Page	Tested in browser
4	Customer create	/staging/forms/customer-create	Page	Tested in browser
5	Customer edit	/staging/customers/{id}/edit	Parameterized page	Tested in browser
6	Product	/staging/forms/product	Page	Tested in browser
7	Order	/staging/forms/order	Page	Tested in browser
8	Order item	/staging/forms/order-item	Page	Tested in browser
9	QA Admin Inventory	/staging/admin	Page	Tested in browser

Counting rule: the dashboard Quick note is a distinct form but not a distinct navigation level because its client-side tab leaves the URL unchanged. The deprecated /staging/legacy/contact route is out of scope. No form was skipped because of the cap; all nine in-scope forms were browser-tested.

Dataset Summary

The Google Sheet contains 103 run-tagged rows. Values are fictional and domain-representative: example.com email addresses, 555-01xx fictional telephone numbers, sample/test addresses, QA-prefixed SKUs and order references, realistic decimal prices and stock quantities, and identifiers that preserve declared entity relationships.

Category	Rows	Purpose
Happy path	42	Valid representative values and composite records
Boundary	9	Minimum, maximum, empty-optional, and size-limit cases
Negative	45	Invalid-format, missing, range, uniqueness, relationship, and injection probes
Regression	1	Closed-duplicate candidate for phone-format bypass
UAT	6	End-to-end composite payloads preserving customer/product/order dependencies
Total	103	Every declared field represented; executed safe negatives carry observed outcomes

Dataset sheet: [📄 Test Data Repository - sample-qa-test-project](#)

Every row carries the run identifier, module, form, route, field/payload, scenario, generated value, expected outcome, validation class, source rule, relationship notes, disposition, and generation date. Executed rows now record observed outcomes, safe retest evidence, severity, and Jira keys; unselected reusable scenarios remain clearly marked as not executed.

Validation Rules

Form / entity	Field group	Rule under review
Registration	username	3-40 characters; begins with a letter; letters/digits/_/- only
Registration	email	Email format; uniqueness intended
Registration	display_name	Optional; maximum 80
Registration	role	qa_viewer or qa_editor
Customer	full_name	2-100 characters
Customer	email	Email format; uniqueness intended
Customer	phone	Optional; maximum 30; formatted phone intended
Customer	address fields	Address 1: 3-120; Address 2: optional <=120; city 2-80; postal code 3-20
Customer	notes/status	Notes optional <=1000; status constrained by form
Product	sku/name	SKU 3-24 and normalized; name 2-120; uniqueness intended for SKU
Product	price/stock	Price intended positive, <=999999.99, two decimals; stock 0-1,000,000
Product	description/category/active	Description optional <=1000; form category; availability boolean
Order	customer/product/quantity	Positive related IDs; quantity 1-100 and not over stock
Order	shipping	Address required 1-200; method standard/express/pickup; gift message optional <=240
Order item	order/product/quantity/price	Valid related IDs; quantity positive and within stock; unit price nonnegative; unique order-product pair
Admin inventory	product/adjustment/reason	Existing product; signed adjustment must not produce invalid stock; reason optional

Relationship Mapping

Parent	Child	Cardinality / constraint	Dataset treatment
customers	orders	One-to-many	Composite UAT payloads create/reference customer before order
orders	order_items	One-to-many; cascade delete	Order item rows reference an existing run order
products	order_items	One-to-many	Valid cases reference run products; negative cases isolate missing IDs
orders + products	order_items	Unique (order_id, product_id)	Duplicate-pair negative case is separate
user_registrations	-	Independent sample records	No authentication semantics inferred
environment_state	-	One fixture-state record	Used only to verify staging identity and seed timestamp

Generated identifiers are run-tagged to avoid collisions. Valid composite payloads preserve dependency order. Invalid relationship probes are deliberately isolated and marked manual-review so they cannot be mistaken for seed/setup data.

Validation Gaps

Six validation gaps were confirmed by browser submission, observation, and an identical safe retest. GAP-03 was deliberately not submitted because the requested safety constraints prohibit creating orphan records. CONTROL-01 produced one temporary endpoint error, then passed on the identical retry and is therefore not a defect.

Oracle ID	Surface	Candidate behavior	Expected severity	This run
GAP-01	Order / shipping address	Required-field bypass	High	Confirmed and retested

Oracle ID	Surface	Candidate behavior	Expected severity	This run
GAP-02	Customer create + edit / phone	Format-rule bypass; no matching open duplicate found	High	Confirmed and retested
GAP-03	Order Item / nonexistent product	Referential-integrity bypass	Critical	Manual review only — not executed (orphan prohibited)
GAP-04	Product / inert reflected markup	Stored/reflected injection	Critical	Confirmed and retested
GAP-05	Order Item / quantity	Zero, negative, or over-stock business-rule bypass	Medium	Confirmed and retested
GAP-06	Product / unit price	ORM validator bypass	High	Confirmed and retested
GAP-07	Registration vs Customer edit / email	Error/status inconsistency	Low	Confirmed and retested
CONTR OL-01	Dashboard Quick note	First 503 then identical retry succeeds	Not a defect	Passed on identical retry — not a defect

Jira Bugs created after duplicate search: TCW-76 (shipping), TCW-77 (phone), TCW-78 (markup reflection), TCW-79 (negative quantity), TCW-80 (negative price), and TCW-81 (email response inconsistency). CONTROL-01 passed its retry and generated no issue.

Edge Cases Generated

- Required fields: omitted, empty string, and whitespace-only values.
- Text bounds: exact minimum/maximum, one below/above, long notes/descriptions, and Unicode-safe representative text.
- Identifiers: zero, negative, nonexistent positive IDs, valid cross-entity IDs, and duplicate order-product pairs.
- Numeric values: zero, negative, decimal precision, upper bounds, over-stock quantities, and signed inventory adjustments.
- Uniqueness: duplicate registration/customer emails and duplicate/normalized SKUs.
- Enumerations and toggles: unsupported roles, methods, statuses, categories, and active/availability states.
- Security-safe rendering: inert markup only; no script execution and no external network contact.
- Failure-path control: Quick note first 503 followed by an identical retry, recorded as non-defect when retry succeeds.

Form Validations that have Failed

Six form-validation failures were reproduced in staging. Each confirmed finding below was submitted twice with the same safe trigger. Screenshots were captured during execution; the Jira connector did not expose an attachment action, so each Bug records the exact route, input, expected result, and observed result.

Finding	Evidence	Severity / Jira
GAP-01	Order shipping_address accepted an empty value twice and persisted both orders.	High — TCW-76
GAP-02 and GAP-07	Invalid customer phone persisted on create and edit; invalid email produced inconsistent registration vs customer-edit responses.	High + Low — TCW-77 and TCW-81

Finding	Evidence	Severity / Jira
GAP-04	Inert markup in a product name was stored and reflected as live HTML twice.	Critical — TCW-78
GAP-05	Negative quantity (-7) persisted twice using existing orders and products; no orphan was created.	Medium — TCW-79
GAP-06	Negative product price (-3.25) persisted twice with unique run-tagged SKUs.	High — TCW-80

Recommendations

1. Enforce shipping_address as required consistently in HTML, request parsing, and persistence; add regression coverage for empty and whitespace-only submissions.
2. Apply one shared phone-format validator to customer creation and edit paths, and add parity tests for both routes.
3. Escape product names and other user-controlled fields at render time; preserve the stored value as text and add an inert-markup regression test.
4. Enforce positive price and quantity constraints in the schema, ORM, and database layers; retain business-rule tests for stock limits and composite uniqueness.
5. Normalize invalid-email handling across registration and customer edit so message structure and status behavior are consistent and user-safe.
6. Keep GAP-03 as a controlled database-integrity review. Test it only in a disposable fixture where cleanup is explicitly authorized and verify that no orphan can remain.

Future Datasets Suggestions

- Locale variants for international names, addresses, postal formats, decimal presentation, and non-ASCII display text.
- State-transition matrices for customer/order statuses and inventory adjustments across successive submissions.
- Concurrency cases for SKU/email uniqueness and duplicate order-product insertion.
- Large but safe order graphs with several products to verify capture-price behavior and cascade deletion.
- Accessibility-oriented field labels, keyboard sequence, error focus, and screen-reader announcement checks.
- API/UI parity datasets that compare Pydantic validation, ORM validators, database constraints, and rendered form feedback.

Run Disposition

This run completed with findings. Repository analysis, scope discovery, deterministic data generation, staging verification, browser execution, identical retesting, Sheet publication, Jira creation, report update, and Teams notification were completed. Seventeen run-tagged QA records remain as staging evidence; all declared relationships are intact and the database contains zero orphans.

Disposition item	Value
Run identifier	sample-qa-test-project-20260805-113952
Workflow completion	Completed through reporting and notification
Forms browser-tested	9
Cap-skipped forms	0
Staging records created	17
Confirmed gaps / Jira bugs	6 / 6
Required next condition	Completed; follow-up tracked in TCW-76 through TCW-81